

Exact Rational Bidisk Lifts for Matrix Laurent Binomials on an Annulus

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Status and source question

Status: candidate partial result, likely valid; human review requested.

Let

$$A_{R,r} = \{z \in \mathbb{C} : r \leq |z| \leq R\}, \quad 0 < r < R, \quad \gamma(z) = \left(\frac{z}{R}, \frac{r}{z}\right).$$

Baran, Pikul, Woerdeman, and Wojtylak prove in Corollary 5.2 that for every rational matrix function $F \in \mathcal{M}_n(\mathcal{R}(A_{R,r}))$,

$$\|F\|_{\mathcal{T}_{A_{R,r}}} = \inf \left\{ \|G\|_{\infty, \mathbb{D}^2} : G \in \mathcal{M}_n(\mathcal{R}(\overline{\mathbb{D}^2})), G \circ \gamma = F \right\}. \quad (1)$$

They then ask for which functions the infimum is attained, saying that the general question is unanswered, and prove a scalar mixed-binomial special case in Proposition 5.3 [1]. The source passage is reproduced below.

Theorem 2.6 gives us the following corollary, cf. Mittal [35] for an analogous result for H^∞ functions supported by an extensive theory.

Corollary 5.2. *For all $F \in \mathcal{M}_n(\mathcal{R}(A_{R,r}))$ we have the following identity for the Agler norm on the annulus:*

$$(31) \quad \|F\|_{\mathcal{T}_{A_{R,r}}} = \inf \left\{ \sup_{z \in A_{R,r}} \|G(z)\| : G \in \mathcal{M}_n(\mathcal{R}(\overline{\mathbb{D}^2})), \gamma^*(G) = F \right\}.$$

Proof. For $F \in \mathcal{M}_n(\mathcal{R}(A_{R,r}))$, due to Theorem 2.6, we have

$$\|F\|_{\mathcal{T}_{\ker \gamma^*}} = \|F\|_{\mathcal{T}_{A_{R,r}}}$$

According to Ando's theorem, this implies (31). $\square$

It is natural to ask for which functions the infimum in (31) is attained. While in general it is not yet answered, we discuss some special cases. For clarity, in the following proposition we restrict our attention to $A_{1,r}$. It is worth noting that using the well known fact, that two annuli are conformally equivalent if and only if their radii have the same ratio. Hence, considerations for $A_{R,r}$ are equivalent to those for $A_{1,\frac{r}{R}}$.

Proposition 5.3. *Let $F(z) = z^k + \frac{1}{z^l}$ where k and l are positive integers and let $G(z_1, z_2) = z_1^k + \frac{z_2^l}{r^l}$. Then*

$$\|F\|_{\mathcal{T}_{A_{1,r}}} = \|G\|_\infty := \sup_{z \in \overline{\mathbb{D}^2}} \|G(z)\|$$

The displayed formula in the source screenshot writes its supremum variable as though G were one-variable. Formula (1) uses the bidisk supremum intended by the preceding theorem and used explicitly in the source's Proposition 5.3.

For an invertible Hilbert-space operator T , membership in the annular class used here is simply

$$T \in \mathcal{T}_{A_{R,r}} \iff \|T\| \leq R \quad \text{and} \quad \|T^{-1}\| \leq r^{-1}. \quad (2)$$

For a matrix-valued function, $F(T)$ acts on the tensor product with $\mathbb{C}^n$; thus, for example, $F(z) = z^p A + z^q B$ gives $F(T) = T^p \otimes A + T^q \otimes B$.

Idea of the proof

The upper bound comes from an explicit bidisk lift. For mixed powers $p = k > 0 > q = -\ell$, the natural lift is

$$G(z_1, z_2) = R^k z_1^k A + r^{-\ell} z_2^\ell B.$$

Its bidisk norm is the maximum of $\|R^k A + \omega r^{-\ell} B\|$ over one relative phase $|\omega| = 1$.

The key point is to recover every such phase from an annular operator. Take a cyclic weighted shift of size $k + \ell$, with k successive weights R and ℓ successive weights r . It has norm R and inverse norm r^{-1} . Starting from one carefully chosen coordinate, both T^k and $T^{-\ell}$ arrive at the same output coordinate, with respective weights R^k and $r^{-\ell}$. Replacing T by αT changes their relative phase by $\alpha^{-(k+\ell)}$, which sweeps the full unit circle. This one compression therefore matches the explicit lift norm even for arbitrary matrix coefficients. Same-sign powers need only scalar operators on the appropriate boundary circle.

Exact Laurent-binomial theorem

Theorem 1. *Let $0 < r < R$, let $p > q$ be integers, and let $A, B \in \mathcal{M}_n(\mathbb{C})$. Put*

$$F(z) = z^p A + z^q B.$$

Define the positive amplitudes a_p, a_q by

$$(a_p, a_q) = \begin{cases} (R^p, R^q), & q \geq 0, \\ (r^p, r^q), & p \leq 0, \\ (R^p, r^q), & p > 0 > q. \end{cases}$$

Then

$$\boxed{\|F\|_{\mathcal{T}_{A_{R,r}}} = \max_{|\omega|=1} \|a_p A + \omega a_q B\|}. \quad (3)$$

In all three cases, the infimum in (1) is attained. Explicit attaining lifts are

$$G(z_1, z_2) = \begin{cases} R^p z_1^p A + R^q z_1^q B, & q \geq 0, \\ r^p z_2^{-p} A + r^q z_2^{-q} B, & p \leq 0, \\ R^p z_1^p A + r^q z_2^{-q} B, & p > 0 > q. \end{cases} \quad (4)$$

We first isolate a standard elementary observation.

Lemma 1. *If G is a matrix polynomial on the bidisk, then*

$$\sup_{(z_1, z_2) \in \mathbb{D}^2} \|G(z_1, z_2)\| = \sup_{|z_1|=|z_2|=1} \|G(z_1, z_2)\|.$$

Consequently, for nonnegative integers u, v and positive a, b ,

$$\|az_1^u A + bz_2^v B\|_{\infty, \mathbb{D}^2} = \max_{|\omega|=1} \|aA + \omega bB\|.$$

The same formula holds when both monomials use the same coordinate and have distinct nonnegative powers.

Proof. For unit vectors $x, y \in \mathbb{C}^n$, the scalar function $y^* G(z_1, z_2) x$ is a polynomial. Applying the maximum-modulus principle successively in the two variables, and then taking the supremum over x, y , proves the first assertion. On the torus, multiplication by the scalar phase of the first monomial does not change the matrix norm, while the quotient of the two monomial phases runs through the whole unit circle. This proves the remaining assertions. The possibility of a zero exponent causes no change. $\square$

Proof of Theorem 1. Every function in (4) is a matrix polynomial and direct substitution of $\gamma(z) = (z/R, r/z)$ gives $G \circ \gamma = F$. For example, in the mixed case $p = k > 0 > q = -\ell$,

$$R^k (z/R)^k A + r^{-\ell} (r/z)^\ell B = z^k A + z^{-\ell} B.$$

Lemma 1 shows in each of the three cases that

$$\|G\|_{\infty, \mathbb{D}^2} = \max_{|\omega|=1} \|a_p A + \omega a_q B\|. \quad (5)$$

Therefore (1) gives the upper inequality in (3).

Suppose first that $q \geq 0$. For every $|\xi| = 1$, the scalar operator $T = R\xi I$ belongs to $\mathcal{T}_{A_R, r}$, and

$$F(T) = \xi^p (R^p A + \xi^{q-p} R^q B).$$

Because ξ^{q-p} runs through the unit circle, the definition of the Agler norm gives the reverse of (5). If $p \leq 0$, use instead $T = r\xi I$. It again lies in the annular class, and exactly the same argument gives the reverse inequality with amplitudes r^p, r^q .

It remains to treat $p = k > 0 > q = -\ell$. Set $N = k + \ell$, and let $e_1, \dots, e_N$ be the standard basis of $\mathbb{C}^N$. Define the cyclic weighted backward shift T by

$$Te_{j+1} = w_j e_j \quad (1 \leq j < N), \quad Te_1 = w_N e_N, \quad (6)$$

where

$$w_1 = \dots = w_k = R, \quad w_{k+1} = \dots = w_N = r.$$

The columns of T are orthogonal and have lengths $w_N, w_1, \dots, w_{N-1}$. Hence

$$\|T\| = R, \quad \|T^{-1}\| = r^{-1},$$

so $T \in \mathcal{T}_{A_R, r}$. Following the two directed paths from e_{k+1} in (6) gives

$$T^k e_{k+1} = R^k e_1, \quad T^{-\ell} e_{k+1} = r^{-\ell} e_1. \quad (7)$$

For $|\alpha| = 1$, also $\alpha T \in \mathcal{T}_{A_{R,r}}$. Thus, for every $x \in \mathbb{C}^n$, (7) yields

$$\begin{aligned} F(\alpha T)(e_{k+1} \otimes x) &= e_1 \otimes (\alpha^k R^k A + \alpha^{-\ell} r^{-\ell} B)x \\ &= \alpha^k e_1 \otimes (R^k A + \alpha^{-(k+\ell)} r^{-\ell} B)x. \end{aligned}$$

Taking the supremum over unit x , then over α , gives

$$\|F\|_{\mathcal{T}_{A_{R,r}}} \geq \max_{|\omega|=1} \|R^k A + \omega r^{-\ell} B\|,$$

because $\alpha^{-(k+\ell)}$ covers the full unit circle. This matches (5), completing both the norm identity and attainment. $\square$

Two broader one-sided classes

Proposition 1. *The infimum in (1) is attained in either of the following situations.*

1. *H is a rational matrix function analytic on a neighbourhood of $\{z : |z| \leq R\}$, and $F = H|_{A_{R,r}}$. An attaining lift is $G(z_1, z_2) = H(Rz_1)$.*
2. *K is a rational matrix function analytic on a neighbourhood of $\overline{\mathbb{D}}$, and $F(z) = K(r/z)$. An attaining lift is $G(z_1, z_2) = K(z_2)$.*

In the first case the exact norm is $\max_{|z|=R} \|H(z)\|$; in the second it is $\max_{|w|=1} \|K(w)\|$.

Proof. The proposed functions are admissible rational lifts. Their bidisk norms are the stated boundary maxima by the scalar maximum-modulus argument used in Lemma 1. Conversely, the scalar operators $T = R\xi I$ in the first case and $T = r\xi I$ in the second case belong to the annular class and recover every point on the relevant boundary circle. The definition of the Agler norm supplies matching lower bounds. $\square$

Verification notes

The proof is exact and does not depend on computation. The script `code/verify_weighted_shift.py` constructs the shift (6) for four representative choices of R, r, k, ℓ, n . It checks its largest and smallest singular values, the inverse-norm constraint, the block compression in (7), the phase identity, and the restriction $G \circ \gamma = F$. The command used was

```
conda run --no-capture-output -n sandbox python code/verify_weighted_shift.py
```

These are transcription checks, not a replacement for the proof.

Scope and full-classification upgrade attempts

Theorem 1 classifies every two-term Laurent support, all matrix sizes, and all annulus radii. It does *not* classify arbitrary rational matrix functions. In particular, it does not decide attainment for the source's example $z + z^{-1} + 1$ on $A_{1,1/2}$.

Eight focused upgrade attempts were made. Montel and weak-star compactness can produce bounded holomorphic limiting lifts but do not preserve rationality or neighbourhood analyticity. Finite-dimensional realizations give no uniform state-dimension bound along a minimizing sequence.

A quotient-ideal counterexample scheme would require proving that a rational coset has only a nonrational best H^∞ representative, for which no usable uniqueness or nonproximality criterion was found. Polynomial-correction optimization for the affine source example was exploratory only. A small exact annular operator gives a lower value near 3.549, but no matching rational lift was proved. The successful scalar-coefficient, matrix-coefficient, arbitrary- radius, same-sign, and one-sided upgrades culminate in the results above.

The obstacle to a mechanical extension to three or more mixed Laurent terms is concrete: the cyclic path aligns two powers and supplies one free relative phase, while three arbitrary matrix coefficients generally require more simultaneous phase information. With the compactness and quotient routes also blocked by the absence of a rational-degree bound, no further credible route to the full classification remained in the allotted upgrade budget.

Bounded novelty check

A bounded search through August 11, 2026 covered the run's four lightweight indexes, the local source corpus, and official-arXiv searches using the arXiv identifier, exact title, author names, 'annulus infimum attained,' 'rational norm-preserving extension,' 'matrix Laurent binomial,' 'Agler norm,' and later papers citing the source. No duplicate packet, later explicit classification, or matching matrix Laurent-binomial theorem was found. This is only a bounded novelty statement, not a claim of priority.

Human-review recommendation

Check the orientation of the weighted shift and both paths in (7); confirm that the matrix-valued maximum-modulus reduction in Lemma 1 loses no information; and check the scalar boundary operators in the same-sign and one-sided cases. The limitation to two-term mixed Laurent support should remain explicit.

References

- [1] R. Baran, P. Pikul, H. J. Woerdeman, and M. Wojtylak, *Contractive realization theory for the annulus and other intersections of discs on the Riemann sphere*, arXiv:2504.03236 (2025), especially Corollary 5.2 and Proposition 5.3.